

4 (a) (i) zero field (strength) inside spheres B1 [1]

(ii) *either* field strength is zero
or the fields are in opposite directions
at a point between the spheres M1
A1 [2]

(b) (i) field strength is (–) potential gradient (*not* V/x) B1 [1]

(ii) 1. field strength has maximum value B1
at $x = 11.4$ cm B1 [2]

2. field strength is zero B1
either at $x = 7.9$ cm (*allow* ± 0.3 cm)
or at 0 to 1.4 cm *or* 11.4 cm to 12 cm B1 [2]

5 (a) (i) $B_{av}(\sin \theta)$ or $B_{av}(\cos \theta)$ B1 [1]